

configure it completely from a Web-based GUI using your browser. There is no need to tweak or configure config files manually.

- **Distributed builds:** Hudson can distribute build/test loads to multiple computers—it uses hardware resources effectively.
- **JUnit/TestNG test reporting:** The Hudson system can parse the JUnit/TestNG test output generated by build/test scripts and generate the intuitive report.
- **Plug-ins:** You can extend its functionality by using various plug-ins. You can write your own plug-ins to support the tools and processes you use, or visit the following link to view all available Hudson plug-ins: wiki.hudson-ci.org/display/HUDSON/Plugins

Jobs

Builds in Hudson are configured as 'Jobs'. They may involve one or many tasks, including code compilation, executing unit tests, publishing build artefacts, updating issue trackers, identifying code violations, generating documentation, etc.

The build can be configured as a series of shell, Perl and Python commands. The best method is to write a script using any scripting language that performs all activities and execute the script from a Hudson job. Depending on the plug-ins installed, you may see many options in the 'job configuration' screen. The job may belong to any of following categories:

- **Scheduled** – The job can be scheduled to run at a specific date/time—*crontab*-like scheduling is possible.
- **Monitor VCS** – You can configure a job to run immediately after a commit in the VCS. The Hudson system can poll the VCS periodically, detect the changes if any, and initiate the build process.
- **On demand** – A build can be triggered on a demand-basis.

Figure 3 demonstrates the Job Dashboard. As you can see, it displays details about the job—including test trend, disk usage report and violations (if any).

In short, you can configure a job to update the VCS working copy, execute the build, generate the installation packages, run tests on different platforms, generate documentation and code quality reports. The final result is published in the user interface.

Plug-ins

You can manage the various plug-ins you require from within the user interface. All users have the ability to keep track of installed and new plug-ins. In fact, you can track the updates to Hudson and the plug-ins from the same screen. Users are notified of updates automatically.

Here are some of the tasks for which you might want to consider the use of plug-ins:

- Leader board
- Report code violation
- Webtest plug-in
- Publish documents
- Integration with FTP/SCP servers
- Integration with issue trackers

You can configure Hudson to run multiple slave instances in parallel, on different platforms. We will discuss the Hudson master-slave set-up in the following section.

Master-slave set-up

You can schedule your tasks to run on the same machine (master), or on a different machine (slave). A master's job is to manage one or more slaves. However, remember that although a master serves HTTP requests, it can still build projects on its own.

Slaves are computers that are set up to build projects triggered by the master. A separate program called a 'slave agent' runs on a slave computer. Let us now understand how to set up Hudson to execute distributed builds using a typical master-slave set-up. Note that you can configure a single


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Regd. Off: Wing 1 & 2, IV Floor,
 Jhaver Plaza, 1A, N.H. Road,
 Nungambakkam, Chennai - 34.
Ph : 28263529 / 3530 / 3540
Telefax : 28263527
E-mail : enquiry@vectratech.in
www.vectratech.in